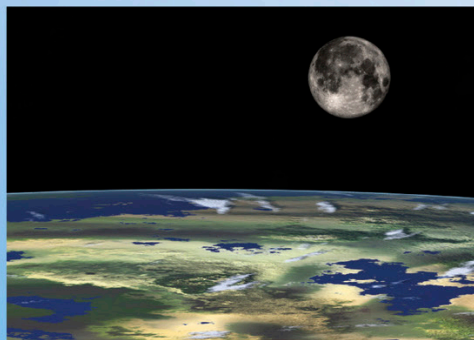




Gravity

The force that makes things fall to the ground is gravity. Gravity keeps the Earth in orbit around the Sun and keeps the Moon in orbit around the Earth. It is one of the most important forces in the universe.

Forces

A force is simply a push or a pull. When you push or pull something to make it move, you are using forces. Some forces work only when objects are touching, but others, such as gravity and magnetism, work at a distance.

This NASA space shuttle, which was in operation until 2011, needed three rockets to help it escape from the Earth's gravitational pull.



Lift-off

A huge force is needed to make a spacecraft take off and escape the Earth's gravity. A force called thrust is provided by rockets. The rockets make hot gases, which expand and stream out at the bottom to push the spacecraft up into the air at great speed.

In a spin

On a merry-go-round, the riders feel they're being pushed outward. This pushing, called centrifugal force, isn't a real force. It's caused by the riders' bodies trying to move in a straight line while the chains are holding them back.

